GERMÁN KÜBER

Senior Backend Engineer — Production Python — Real-Time AI & Voice Pipelines

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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PROFESSIONAL SUMMARY

Senior backend engineer with 15+ years building production systems and a deep focus over the last several years on production Python (FastAPI, Django, Flask) for real-time AI products. Shipped a real-time audio AI pipeline at KliverAI on OpenAI Realtime API with sub-200ms streaming latency and a WebSocket-based transcription pipeline capturing live meeting audio — directly applicable to ambient AI / scribe products. Operated production Python and Rust services on Docker and Kubernetes / AWS at high scale (Borderless Capital: $50M+ daily transactional flow). Comfortable across adjacent stacks including **TypeScript / Node.js** and .NET. Customer-facing impact orientation; comfortable with ambiguity and fast iteration; very much a fit for ship-it-now environments. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

CORE EXPERTISE

Production Python: FastAPI, Flask, Django, asyncio, async / await, Cython, PyO3 bindings, Pydantic, SQLAlchemy, REST and WebSocket APIs, background workers, structured logging, testing (pytest)

Real-Time AI & Voice Pipelines: OpenAI Realtime API, WebSocket streaming, SignalR, sub-200ms audio pipelines, real-time transcription capture, real-time evaluation pipelines, LLM orchestration (OpenAI, Anthropic, Gemini), multi-provider routing, prompt engineering, RAG, multi-agent (LangGraph), MCP server / client

Cloud & Infrastructure: AWS, Docker, Kubernetes (manifests, deployments, services), CI / CD pipelines, Linux, Infrastructure as Code (Terraform), observability stacks

Data & Storage: PostgreSQL (OLTP, jsonb, pgvector, semantic cache backing store), Neo4j (Cypher), caching and pub/sub (Redis-style), high-throughput data pipelines

Backend Engineering: Event-driven microservices, async coordination, fault tolerance, robust APIs and abstractions, Domain-Driven Design, customer-facing systems at scale

Agent Engineering: Production AI agents, agentic workflows, LangChain, LangGraph, Microsoft Agent Framework, A2A protocol, eval frameworks (LangSmith, Langfuse, RAGAS, DeepEval), context engineering, knowledge graphs (Neo4j, Cognee, Graphiti), ontology design, code graphs, MCP servers, LLM-as-judge evals, LoRA fine-tuning, semantic caching, human-in-the-loop

Paradigms, Design & Practices: OOP, Functional Programming, Design Patterns, Architecture Patterns, SOLID, data structures, algorithm design, clean code, Test-Driven Development (TDD), pair programming, code review, mentoring, ADRs, CI / CD, Infrastructure as Code (Terraform)

Observability & Reliability: Prometheus, OpenTelemetry, distributed tracing, structured logging, 50+ production metrics, 24 / 7 on-call rotation, post-mortems, incident response

Adjacent Stacks: TypeScript / Node.js, Rust (tokio, async), .NET / C#, Go (reverse-engineering Geth internals)

PROFESSIONAL EXPERIENCE

AI Architect

Nera — Feb 2026 – Present (full-time) — Remote

- Own Nera's AI architecture in Python: the knowledge and code-intelligence layer that lets the whole company query the business in natural language, and the agentic tooling product and engineering use to go from problem to spec.
- Built an AI product-engineering platform where multi-agent LangGraph flows take a problem through PAD, PRD, epic and tech spec, grounded in the code graph and business knowledge.
- Designed and built a company-wide ontology (28 classes, 26 invariants) with ingestion pipelines from Notion, Jira and GitHub into a Neo4j knowledge graph plus a pgvector semantic index, giving every area end-to-end traceability from problem to PR and natural-language Q&A over MCP.
- Built a code graph of the entire microservice fleet (66 repos, 77k nodes, 210k relationships across Java, TypeScript and Go), enriched with runtime calls from OpenTelemetry and served through a 32-tool MCP server that engineers use to design features and product uses to write PRDs.
- Automated user-research capture with a pipeline that transcribes interviews and extracts verifiable quotes and insights into Notion with human approval, replacing a manual tracker.
- Made evals a release gate: record-and-replay harnesses, a 100-question benchmark for the knowledge graph, fidelity audits for the code graph and LLM-as-judge for generated PRDs.
- Drove Spec-Driven Development (SDD) across engineering with a shared framework (specs, TDD, review gates) that standardized how features are built and raised quality team-wide.

Independent AI Consultant — Agent Engineering & Enterprise AI Delivery

Self-employed (autónomo, Spain) — Mar 2026 – Jul 2026 — Remote, client under NDA

- Forward-deployed AI consultant embedded with an enterprise engineering organization: work directly with the customer to run discovery with business and technical stakeholders, architect the solution, and own delivery from pilot through production and continuous iteration.
- Design and ship production agentic workflows in Python on LangChain, LangGraph and Microsoft Agent Framework, testing each graph, running eval frameworks and eval suites against it, and monitoring it live in production.
- Own the RAG platform end-to-end for internal company documentation — retrieval optimization, chunking, context engineering, and regression test suites gating every pipeline change.
- Fine-tuned models with LoRA on internal company documents deployed through Azure AI Foundry; cut cost and latency through semantic caching, semantic memory, and dynamic model routing.

Lead AI Architect & Co-Founder

KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain

- Designed and shipped a real-time audio AI pipeline on OpenAI Realtime API with SignalR streaming, achieving sub-200ms end-to-end latency across concurrent user sessions — capturing live conversation, processing it through LLMs, and returning structured output in real time.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms latency.
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.
- Built a real-time transcription capture pipeline: a Chrome Extension streams live Google Meet audio over WebSocket into a Python backend, which then feeds an automated AI evaluation pipeline.
- Built backend services in Python (FastAPI, asyncio) for AI integrations and data pipelines, alongside Node.js / TypeScript services for the API and streaming layer; deployed everything on Docker and Kubernetes on AWS.
- Integrated three LLM providers (OpenAI, Anthropic, Gemini) with dynamic model routing, picking the best model per task to cut inference spend.
- Built an adaptive multi-agent platform on LangGraph with multi-step branching logic and RAG-powered personalization.

- Implemented MCP (Model Context Protocol) servers and clients to standardize tool integration across services.
- Established the observability stack with OpenTelemetry, distributed tracing, structured logging, and LLM-specific tracing (Langfuse, LangSmith) plus prompt evaluation (Braintrust, PromptLayer, Promptfoo).
- Hired and led a distributed team of up to 3 engineers, owning technical interviews, hiring, and architectural reviews — hands-on in the codebase throughout.

Fractional CTO (Part-time, hands-on)

Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Shipped customer-facing AI features in production using Python (FastAPI, LangChain) and Node.js / TypeScript, establishing the team's first LLM evaluation and monitoring practices.
- Optimized LLM API usage and cloud infrastructure (AWS, Docker, Kubernetes) through model selection, prompt compression, and semantic caching — cutting combined LLM API and cloud spend by 45%.
- Designed a reusable base architecture on LangChain and LangGraph with Langfuse / LangSmith monitoring; introduced Spec-Driven Development (SDD); proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET.

Lead Backend Engineer (Distributed Systems, Python & Rust)

Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote

- Operated production backend services processing 10K+ opportunities per day from $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L with execution under 10ms p99.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Built Python services (endpoints, business logic, external integrations) running on Docker and Kubernetes in production, alongside the Rust core; owned K8s manifests and deployment configuration for the services I shipped.
- Built Python / Cython and PyO3 FFI bindings exposing Rust performance-critical code to higher-level Python orchestration layers, eliminating GC pauses across language boundaries.
- Used Postgres as the system of record for pipeline state, opportunities, and execution outcomes, and Redis-style caching both for low-latency caching and as a pub / sub backbone between services.
- Engineered low-latency observability with Prometheus and custom telemetry, tracking 50+ real-time metrics across the production pipeline.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Owned 24 / 7 production reliability and incident response; acted as technical lead across the blockchain team, which grew from 4 to 10 engineers — code review, mentoring, and hiring.

Senior Software Architect (Backend, Event-Driven)

Paramo Technology — Oct 2020 – Sep 2021 — Remote

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed event-driven microservices in .NET, implemented distributed tracing (OpenTelemetry, Application Insights, Datadog), brought in Kubernetes to scale, and built automated deploy pipelines.
- Led part of the hiring and designed the technical entry exam; delivered internal training on architecture, design patterns, and operational practices.

Software Architect

CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)

- Was the company's first employee; led architecture decisions across several enterprise product lines, designing microservices backends with Domain-Driven Design.
- Built a distributed system over geospatial data with PostGIS for a UK client; delivered technical training and conference talks on backend architecture.

Software Architect / Team Lead — Earlier Roles

Majestic Hotel Group (2014–2017) · Plus54 (2011–2014) · Acsys (2009–2011) · IBM (2008–2009) — Argentina

- Full-stack features for internal hotel management platforms, first as Team Lead then architect; led migration from AngularJS to Angular 2 and redesigned the data model toward event-oriented architecture. Earlier: .NET web applications, internal tooling and mobile components end to end.

KEY ACHIEVEMENTS

Real-time audio AI pipeline at KliverAI — OpenAI Realtime API + SignalR + WebSocket transcription capture, sub-200ms streaming latency across concurrent sessions.

Production backend at Borderless — Python and Rust services on Docker / Kubernetes / AWS, processing $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L at sub-10ms p99.

Company-Wide Knowledge & Code Intelligence at Nera — 28-class ontology over Notion, Jira and GitHub in Neo4j + pgvector, plus a 66-repo / 77k-node code graph served through a 32-tool MCP server, with a 100-question benchmark and LLM-as-judge evals gating every release.

Production AI agents (Independent Consulting) — Agentic workflows in Python on LangChain, LangGraph and Microsoft Agent Framework, eval-gated and monitored live in production for an enterprise client.

Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present). **80+ International Technical Talks** — conferences and meetups across LATAM and Europe. **Net-Baires Founder** — Argentina's largest .NET developer community, 100K+ members across 10+ years.

EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) · Azure Associate (AZ-104 / AZ-204 level) — Microsoft
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)
Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

OPEN SOURCE, WRITING & COMMUNITY

YouTube channel (youtube.com/@germankuber) — technical deep-dives on AI, Python, Rust, and software architecture.
Net-Baires (.NET Community) — Founder (2015–present), 100K+ members. **Crypto Para Mortales** — Founder (2023–present), Spanish-language blockchain education conference. **Patrones Para Mortales** — Founder (2020–present), educational content on software design.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams